PHASE 4 – PROJECT PLANNING

Automated Standard Laptop Procurement System

Project Planning defines how the project will be organized, developed, tested, documented, and demonstrated. This phase converts the identified requirements into a practical development plan. It also assigns responsibilities to team members, defines the development sequence, identifies expected risks, and provides methods to control those risks.

1. Project Goal

The main goal of the project is to develop an automated system that simplifies and streamlines the standard laptop procurement process. The system is intended to reduce manual work, improve the accuracy of procurement requests, provide a clear workflow, and make the overall process easier to manage.

The project aims to:

- Automate repetitive activities involved in standard laptop procurement.
- Reduce manual data entry and minimize human errors.
- Provide a structured process for submitting and processing laptop requests.
- Validate request information before processing.
- Improve communication and tracking of procurement activities.
- Reduce processing time by following a predefined workflow.
- Maintain proper records of requests and their processing status.
- Provide a simple and user-friendly process for the concerned users.
- Support testing, error correction, documentation, and final demonstration.

2. Work Distribution

Work distribution is important because each team member is assigned a specific responsibility. This avoids confusion, improves coordination, and helps the project progress in parallel.

Team Member	Responsibility	Major Activities
Member 1	Project Design	Prepare workflow, system structure, process flow, and design s ecifications.
Member 2	Development	Create system components, configure automation, and implem nt the planned
Member 3	Testing	Prepare test cases, test system functions, identify errors, and v rify corrections.
Member 4	Documentation & Demo	Prepare project document, screenshots, presentation material, and final demons

3. Development Plan

The development plan describes the activities that will be completed from the beginning of the project until the final demonstration. Each step has a specific purpose and should be completed in the correct sequence.

Step 1: Understand the Existing Procurement Problem

The team studies the existing standard laptop procurement process and identifies problems such as manual request handling, incorrect information, repeated data entry, delays, configuration mistakes, and difficulty in tracking requests. The existing workflow is documented so that the proposed automated system can address the identified problems.

Step 2: Identify System Requirements

Functional and non-functional requirements are collected. Functional requirements describe what the system should do, such as accepting requests, validating information, processing requests, updating status, and maintaining records. Non-functional requirements include usability, reliability, security, performance, and maintainability.

Step 3: Design the Workflow

A clear workflow is prepared based on the requirements. The team defines the sequence of activities from request creation to completion. Decision points, validation steps, approvals, notifications, and status updates are considered during this stage.

Step 4: Create the Required System Components

The required screens, forms, data fields, workflow elements, automation components, and supporting configurations are prepared. Each component is created according to the approved design so that the complete process can work together.

Step 5: Implement the Automation

The designed workflow is converted into an automated process. Repetitive activities are configured to run automatically wherever possible. Validation rules, status updates, notifications, and other required automation are implemented and connected.

Step 6: Test the System

The completed components are tested individually and together. Test cases are prepared for normal inputs, incorrect inputs, missing data, workflow conditions, permissions, and expected outputs. The objective is to confirm that the system works according to the requirements.

Step 7: Fix Errors

Errors found during testing are recorded and analyzed. Configuration mistakes, incorrect conditions, validation problems, and other issues are corrected. The affected functions are tested again after correction to ensure that the changes work properly.

Step 8: Prepare Documentation

The team prepares the project document containing the project goal, requirements, design, planning, development, testing, results, screenshots, and other required information. The documentation is organized according to the required project phases.

Step 9: Demonstrate the Final Project

The final system is demonstrated to show the complete workflow. The team explains the problem, proposed solution, important features, automation process, testing results, and final output. The

demonstration confirms that the developed system satisfies the planned objectives.

4. Expected Timeline

The project is planned in a sequence so that each stage provides the necessary input for the next stage. The expected flow is shown below:

Phase	Main Activities	Expected Output
Requirement Analysis	Understand problem, collect requirements, identify inputs and outputs.	Requirement specification
Design	Prepare workflow, system structure, data flow, and process design.	Approved system design
Development	Create components and implement required automation.	Working system
Testing	Execute test cases and verify system behavior.	Test results and defect list
Error Correction	Fix identified problems and retest affected functions.	Corrected system
Documentation	Prepare detailed project document and supporting screenshots.	Final project document
Demonstration	Present the completed system and explain the workflow.	Final project demonstration

Flow: Requirement Analysis → Design → Development → Testing → Error Correction → Documentation → Demonstration

5. Risk Management

Risk management is the process of identifying possible problems before or during development and preparing suitable actions to reduce their effect on the project. The following risks are considered for the proposed system.

Risk	Possible Impact	Risk Control / Mitigation
Incorrect request data	Wrong or incomplete procurement information may cause processing errors.	Use required fields, validation, clear instructions, and input checking.
Configuration errors	Incorrect configuration may cause the workflow or automation to behave unexpectedly.	Review configurations carefully and test each component before integration.
Automation failures	Automated steps may not execute as expected, causing delays or incomplete processing.	Test different workflow conditions and verify each automation step.
Access permission issues	Users may be unable to access required functions or may receive unauthorized access.	Check roles, permissions, and access settings during testing.

Risk	Possible Impact	Risk Control / Mitigation
Testing errors	Some defects may remain undetected if testing is incomplete.	Prepare test cases, cover normal and abnormal inputs, and perform retesting after fixes.
Documentation gaps	Important project activities or results may not be properly recorded.	Maintain documentation throughout development and update it after major changes.
Time constraints	Delay in one activity may affect later project stages.	Follow the planned sequence, assign responsibilities clearly, and monitor progress regularly.

6. Quality and Monitoring Plan

Project progress should be monitored throughout the development process. The team should regularly verify whether the planned activities are completed, whether the system behaves as expected, and whether identified issues are resolved.

- Review completed work against the project requirements.
- Check whether assigned responsibilities are completed on time.
- Verify that workflow steps are connected correctly.
- Maintain a record of identified issues and their solutions.
- Repeat testing after important changes.
- Keep screenshots and supporting evidence for documentation.
- Conduct a final review before the project demonstration.

7. Expected Project Outcome

At the end of the project, the team is expected to have a working automated standard laptop procurement system that follows the planned workflow. The system should accept and validate the required information, perform the configured automation, maintain appropriate status or records, and support a clear procurement process. The project should also include completed testing evidence, corrected errors, detailed documentation, and a final demonstration.

8. Conclusion

The Project Planning phase provides a clear roadmap for successful project completion. By dividing the work among team members, following a step-by-step development plan, defining a realistic sequence, and managing possible risks, the team can develop the automated procurement system in an organized manner. Proper planning also helps reduce errors, improve coordination, and ensure that the final project meets its intended objectives.